TEACHING WALLAH

TEACHING Exam

Subject - MATHS

Topic – Algebra

LECTURE NO -02

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Topics to be Covered

Topic Algebra

TEACHING WALLAH

If $a+b=3$ and $a^2+b^2=89$, then What is the value of $\frac{1}{a}+\frac{1}{b}$?

यदि $a+b=3$ और $a^2+b^2=89$ है, तो $\frac{1}{a}+\frac{1}{b}$ का मान क्या होगा?

(a) $\frac{3}{20}$

(b) $-\frac{3}{40}$

(c) $-\frac{1}{4}$

(d) $\frac{1}{8}$

$(a+b)^2=(3)^2$

$a^2+b^2+2ab=9$

$89+2ab=9$

$=9-89$

$\cancel{2}ab=-\cancel{80}$

$ab=-40$

$\frac{b+a}{ab}=\frac{3}{ab}$

$=-\frac{3}{40}$

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If x + y = 3 and $x^2 + y^2 = 29$, then what is the value of $\frac{1}{x} + \frac{1}{y}$?

यदि x + y = 3 और $x^2 + y^2 = 29$ है, तो $\frac{1}{x} + \frac{1}{y}$ का मान है-

$$(x+y)^2 = 3^2$$

$$x^2+y^2+2xy = 9$$

$$29+2xy = 9$$

$$xy = -\cancel{\frac{20}{2}} = -10$$

$$\frac{y+x}{xy} = \frac{3}{xy} = -\frac{3}{10}$$

(a) $\frac{3}{29}$

(b) $-\frac{3}{10}$

(c) $-\frac{1}{5}$

(d) $\frac{3}{10}$

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If $a - b = 7$ and $a^2 + b^2 = 29$, then what is the value of $\frac{1}{a} - \frac{1}{b} = ?$

यदि $a - b = 7$ और $a^2 + b^2 = 29$ है, तो $\frac{1}{a} - \frac{1}{b}$ का मान क्या है?

$-(a-b) = -7$

$b - a = -7$

(a) $\frac{3}{5}$

(b) $-\frac{3}{5}$

(c) $\frac{7}{10}$

(d) $-\frac{7}{20}$

$(a-b)^2 = a^2 + b^2 - 2ab$

$7^2 = 29 - 2ab$

$49 - 29 = -2ab$

$\cancel{20} = -\cancel{2}ab$

$\boxed{-10 = ab}$

$\frac{b-a}{ab} = \frac{-7}{ab}$

$= \frac{+7}{+10}$ ✓

If $x+\frac{1}{x}=3$ thend find

(1) $x^2+\frac{1}{x^2}$.

(2) $x^3+\frac{1}{x^3}$

(3) $x^4+\frac{1}{x^4}$

(4) $x^9+\frac{1}{x^9}$.

(5.) $x^6+\frac{1}{x^6}$.

<u>Solution</u>:- • $x^2+\frac{1}{x^2}=3^2-2=7$ ✓

• $x^3+\frac{1}{x^3}=3^3-3(3)=18$ ✓

• $x^4+\frac{1}{x^4}=7^2-2=47$

• $x^6+\frac{1}{x^6}=18^2-2=322$

• $x^9+\frac{1}{x^9}$

$=18^3-3(18)$

$18[324-3]$

$=321\times18$

$=$ 5778 Ans

If $x+\frac{1}{x}=a$ then find $x^5+\frac{1}{x^5}$?

$$\left(x^2+\frac{1}{x^2}\right)\left(x^3+\frac{1}{x^3}\right) = x^5+\cancel{x^2}\times\frac{1}{\cancel{x^3}}+\frac{1}{\cancel{x^2}}\times\cancel{x^3}+\frac{1}{x^5}$$

$$\left(x^2+\frac{1}{x^2}\right)\left(x^3+\frac{1}{x^3}\right) = \left(x^5+\frac{1}{x^5}\right)+\left(x+\frac{1}{x}\right)$$

$$\boxed{\left(x^2+\frac{1}{x^2}\right)\left(x^3+\frac{1}{x^3}\right) - \left(x+\frac{1}{x}\right) = x^5+\frac{1}{x^5}}$$

$$\boxed{(a^2-2)\times(a^3-3a) - a}$$

If $x+\frac{1}{x}=4$ then find $x^5+\frac{1}{x^5}$?

Soln: → $(4^2-2)\times(4^3-3(4))-4$

$x^5 = x^2 \times x^3$

$= 14\times 52-4$

$= 4(182-1)$

$= 4\times 181 = 724$ Ans

If $x+\frac{1}{x}=a$ $(3+4)$

find $x^7+\frac{1}{x^7}$?

$\left(x^3+\frac{1}{x^3}\right)\times\left(x^4+\frac{1}{x^4}\right)-\left(x+\frac{1}{x}\right)$

cube $\times$ (power)^4 $-$ a

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If $x+\frac{1}{x}=3$ find $x^7+\frac{1}{x^7}$?

cube x four − 3

$\cdot\ 3^2-2=7$

$\cdot\ 7^2-2=47$

$[3^3-3(3)] \times 47-3$

$= 18 \times 47 - 3$

$846 - 3 = 843$

If $x+\frac{1}{x}=4$ then

find $x^5+\frac{1}{x^5}$

Square x cube − 4

$(4^2-2)\times(4^3-4)-4$

$= 14 \times 52 - 4$

$= 724$

→ $x^7+\frac{1}{x^7}$?

↓

cube x four − 4

$= 52 \times (14^2-2) - 4$

$= 52 \times 194 - 4$

$= 4(13 \times 194 - 1)$

$= 2521 \times 4$

$= 10084$ Ans

THANK YOU